

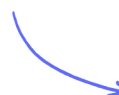
## Structured Questions

# Density, Upthrust & Viscous Drag

Density / Upthrust / Viscous Drag / Core Practical 4: Investigating Viscosity

|                     |            |
|---------------------|------------|
| Medium (1 question) | /7         |
| Hard (2 questions)  | /15        |
| <b>Total Marks</b>  | <b>/22</b> |

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# Medium Questions

- 1 (a) Raindrops of different sizes fall with different terminal velocities through air.

The table shows the measured value of the terminal velocity for raindrops of different sizes.

| Raindrop size | Drop diameter / mm | Terminal velocity / $\text{m s}^{-1}$ |
|---------------|--------------------|---------------------------------------|
| small         | 0.5                | 2.1                                   |
| medium        | 2.0                | 6.5                                   |
| large         | 5.0                | 9.1                                   |

Derive, using Stokes' law, the following expression for the terminal velocity  $v$  of a spherical raindrop in terms of its radius  $r$ .

$$v = \frac{2g\rho r^2}{9\eta}$$

where  $\rho$  is the density of rainwater and  $\eta$  is the viscosity of air.

You should ignore upthrust.

(2 marks)

- (b) Show that the expression given in (a) produces a value of about  $800 \text{ m s}^{-1}$  for the terminal velocity of a large raindrop.

$$\rho = 1.0 \times 10^3 \text{ kg m}^{-3}$$

$$\eta = 1.8 \times 10^{-5} \text{ Pa s}$$

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**(2 marks)**

**(c)** Explain whether Stokes' law is suitable for calculating the terminal velocity of raindrops.

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**(3 marks)**

# Hard Questions

- 1 (a) Genuine crystal balls are made from clarified quartz rather than glass.

A student was given a small crystal ball and wanted to know whether it was genuine.

The mean diameter of the crystal ball was measured to be 5.06 cm and the mass of the crystal ball was measured to be 175 g.

Show that the density of the material of the crystal ball is about  $2600 \text{ kg m}^{-3}$ .

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**(3 marks)**

- (b) The student measured the diameter of the crystal ball using vernier calipers with a resolution of 0.01 cm.

She measured the mass of the crystal ball using a balance with a resolution of 1 g.

The table gives the densities of clarified quartz and glass.

| Material         | Density / $\text{kg m}^{-3}$ |
|------------------|------------------------------|
| Clarified quartz | 2650                         |
| Glass            | 2590                         |

Determine whether the crystal ball was genuine.

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(6 marks)

**2 (a)** A student investigated the terminal velocity of steel spheres falling through oil.

The student obtained the following results.

radius of steel sphere = 1.50 mm

volume of steel sphere =  $1.41 \times 10^{-8} \text{ m}^3$

mass of steel sphere =  $1.10 \times 10^{-4} \text{ kg}$

maximum speed of sphere =  $0.849 \text{ ms}^{-1}$

The student had the following table.

| Type of oil | Density at 26°C / $\text{kg m}^{-3}$ | Viscosity at 26 °C / Pa s |
|-------------|--------------------------------------|---------------------------|
| Corn        | 918                                  | 0.0447                    |
| Hazelnut    | 918                                  | 0.0504                    |
| Sunflower   | 918                                  | 0.0414                    |

Identify which type of oil the student used.

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**(4 marks)**

**(b)** The values in the table are for oil at 26°C.

Explain the effect of carrying out the investigation with oil at a lower temperature.

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(2 marks)